LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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EXP -08

8. Create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Aim: To create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Design:

NETWORK FIELDNOTES / MOBILE SYSTEMS

01 / 02

Public Land Mobile Network (PLMN)

A field guide to the networks that connect mobile subscribers.

02 Roaming explained →

THE BIG IDEA

One operator. Connected services.

A PLMN is a mobile telecommunications network operated by a mobile network operator to provide services to subscribers. It combines radio access and core-network functions to connect users, deliver services and manage mobility.

01 From device to service

A simplified conceptual service path

UE

Mobile Device

Handset or modem + SIM / USIM

RAN

Radio Access Network

Radio sites + applicable radio controllers
BTS/BSC · Node B/RNC · eNodeB · gNodeB

CORE

Core Network

Subscriber records · authentication · mobility · switching / packet sessions

EXTERNAL

Internet / PSTN

Packet-data networks / public switched telephone network

DATA → INTERNET

Packet-core gateways connect data sessions to external packet networks.

VOICE → PSTN, WHERE NEEDED

Legacy MSC switching or IMS and interworking; not the same path as Internet data.

02 The building blocks

Component notes · generation badges indicate applicability

DEVICE

Mobile station / UE

2G 3G 4G 5G

The subscriber's handset or modem and SIM / USIM. UE means user equipment; mobile station is the GSM term.

Uses radio access to request network services.

RADIO ACCESS NETWORK

Base stations & controllers

2G 3G 4G 5G

BTS / Node B provide radio coverage. BSC / RNC control radio resources in GSM / UMTS, respectively.

LTE eNodeB and NR gNodeB have no separate BSC / RNC.

CORE · CIRCUIT SWITCHED

MSC

2G 3G

The Mobile Switching Centre handles legacy circuit-switched voice calls and mobility functions.

Not a universal component of 4G / 5G packet cores.

CORE · SUBSCRIBER RECORDS

HLR / HSS / UDM

2G 3G 4G 5G

Home subscriber profiles and service entitlements: HLR in 2G/3G, HSS in LTE and IMS, UDM in 5G.

Home Location Register / Home Subscriber Server / Unified Data Management.

CORE · SECURITY

Authentication functions

2G 3G 4G 5G

Verify subscriber legitimacy: AuC in legacy networks; HSS/AuC in LTE; AUSF with UDM in 5G.

AuC: Authentication Centre. AUSF: Authentication Server Function.

CORE · PACKET SERVICES

Packet core

2G 3G 4G 5G

Establishes data sessions and forwards packets. 2G/3G: GPRS SGSN/GGSN; 4G: EPC; 5G: 5GC.

Evolved Packet Core / 5G Core. Modern voice uses IMS where deployed.

03 Different generations, different architecture

2G GSM

BTS + BSC

Base Transceiver Station + Base Station Controller

3G UMTS

Node B + RNC

Radio Network Controller manages Node B sites

4G LTE

eNodeB

Radio control integrated; no separate BSC / RNC

5G NR

gNodeB

NR radio access; no separate BSC / RNC

These are alternative generation mappings, not a serial chain. Real deployments can combine generations; 5G non-standalone may use an LTE/EPC anchor.

04 A network has an identity

PLMN ID = MCC + MNC

□ □ □ — MCC · Mobile Country Code, 3 digits

□ □ [□] — MNC · Mobile Network Code, 2 or 3 digits

Digit slots only; brackets mark an optional third MNC digit. Retain leading zeros. A PLMN ID identifies the network—not a subscriber, an IMSI or a phone number.

05 What the network delivers

- **Voice** — Calls via legacy switching or IMS.
- **Messaging** — SMS and supported messaging services.
- **Data** — Packet connectivity for apps and Internet.
- **Mobility management** — Registration, reachability and movement.

Away from home? An authorized visited PLMN can provide access.
Continue to 02: Roaming explained →

READING THE DIAGRAM

2G / 3G / 4G / 5G = technology generation.
→ Conceptual service direction, not a physical interface map.

Educational simplification: functions may be distributed or virtualized. GPRS = General Packet Radio Service; SGSN/GGSN are its serving/gateway support nodes. IMS = IP Multimedia Subsystem. Interfaces and optional elements are omitted.

Remember: radio access connects the device; the core coordinates subscribers and services.

Activate

Roaming: connected beyond home

Two networks cooperate so a subscriber can use services away from their home PLMN.

01 PLMN fundamentals →

SAME SUBSCRIPTION

Different serving network.

The home PLMN (HPLMN) provisions the subscriber. A visited PLMN (VPLMN) supplies local radio access and serving-network functions when roaming agreements and authorization allow it.

01 Local access, home-supported trust

Device physically located in the visited network

VPLMN · VISITED

Where the subscriber is now

Roaming device
Home subscription / SIM remains in use while the device selects a visited PLMN.

Local radio access & registration

Visited RAN + serving functions
Provide radio connectivity, serving mobility and session functions as applicable.

DATA SESSION ——→ Internet

Example only: local breakout via a visited-network data gateway.

Registration / authentication coordination

Home-supported authentication and service authorization

ROAMING AGREEMENT REQUIRED

HPLMN · HOME

Where the subscription belongs

Home subscriber systems
HLR / HSS / UDM, depending on generation: subscription profile and service entitlements.

Home authentication support
AuC, HSS/AuC or AUSF/UDM support the applicable authentication procedures.

Long-term secret keys are not handed to the visited network.

Data may be home-routed or use local breakout, depending on technology and agreement. The solid path above shows local breakout only; voice and messaging paths are not drawn.

Dashed blue ↔ registration / authentication signaling ——→ Solid teal → example user-data direction



Result: -

Hence created an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).